Project: Worked examples
Title: Motor and actuator sizing
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1. Scope and method

Torque, speed and power a motor needs to move a load through a mechanism on a point-to-point cycle: the load reflected to the motor shaft, the acceleration from the motion profile, peak torque during acceleration, RMS torque over the cycle (thermal duty), peak speed, inertia ratio, and a check of a candidate motor's rated and peak torque and speed with a safety margin.

WHAT IT CALCULATES	RELATION
Motor-to-load ratio of a linear mechanism, in metres per motor radian	$k = p/(2\pi\, i)$ (screw); $D/(2\, i)$ (belt); $D_p/(2\, i)$ (rack)
Load inertia reflected to the motor through a linear mechanism	$J_{refl} = m\, k^2/\eta$
Load torque at the motor through a linear mechanism	$T_{load} = F\, k/\eta$, with $F = m\, g\, (\sin\theta + \mu \cos\theta) + F_p$
Reflected inertia and torque of a rotary load through a gearbox	$J_{refl} = J_L/(i^2\, \eta)$; $T_{load} = T_L/(i\, \eta)$
Move profile, trapezoidal 1/3–1/3–1/3 (adjustable)	$v_{pk} = s/(t_{move} - t_{acc})$; $a = v_{pk}/t_{acc}$
Move profile, S-curve (sine acceleration, same peak speed)	$a_{pk} = (\pi/2)\, a_{trap}$
Torque to accelerate the total inertia, and the peak torque	$T_{acc} = J_{total}\, \alpha$; $T_{pk} = T_{acc} + T_{load}$
RMS (thermal) torque over the whole cycle, dwell included	$T_{rms} = \sqrt{(\Sigma\, T_i^2\, t_i / t_{cycle})}$

Basis: Load torque, inertia referred to the motor, duty cycles[1](Ch. on motor and drive selection); Reflected inertia, RMS torque, inertia ratio practice[2].

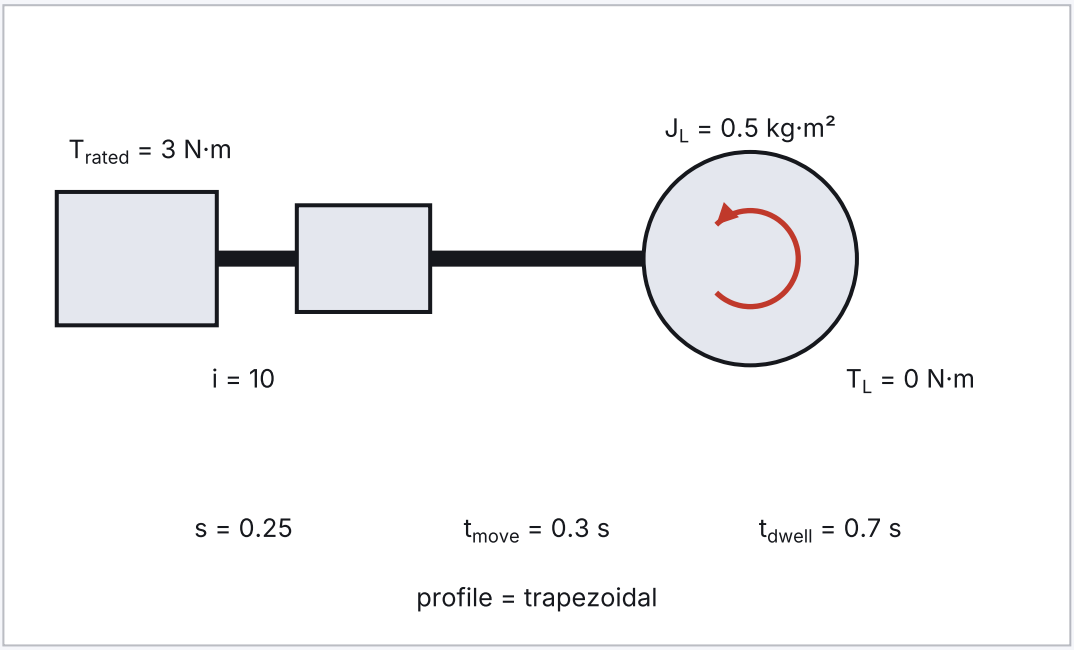


Figure 1 — A motor through a gearbox to a rotary load of inertia J_L and torque T_L. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	gearbox rotary		assumption: Sizing convention.	any
i	Gear ratio i (motor : output, 1 = none)	10		spec: Gearbox datasheet PG-10: 10:1, 85 %	1 … 1000
η	Mechanism efficiency (overall)	0.85		spec: Gearbox datasheet PG-10: 10:1, 85 %	0.3 … 1
J_{mech}	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	0	kg·m²	assumption: Sizing convention.	0 … 10 kg·m²
μ	Friction coefficient (guide / bearing)	0.1		default	0 … 1
θ	Incline from horizontal (+ moving uphill)	0	°	default	-90 … 90 °
F_p	Process force opposing the move (linear)	0	N	default	0 … 1000000 N
J_L	Load inertia at the output (rotary mechanisms)	0.5	kg·m²	spec: S1-style inertia calc of the 600 mm table with fixtures: 0.5 kg·m²	0 … 10000 kg·m²
T_L	Constant load torque at the output (rotary)	0	N·m	assumption: Sizing convention.	0 … 100000 N·m
profile	Motion profile	trapezoidal		assumption: Sizing convention.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
s	Move distance (linear: m; rotary: output revolutions)	0.25		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.0001 … 10000
t_{move}	Move time	0.3	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.01 … 3600 s
f_{acc}	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.05 … 0.5
t_{dwell}	Dwell between moves	0.7	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0 … 3600 s
J_{motor}	Candidate motor rotor inertia	0.0005	kg·m²	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	1e-7 … 10 kg·m²
T_{rated}	Candidate motor continuous (rated) torque	3	N·m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N·m
T_{peak}	Candidate motor peak torque	9	N·m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N·m
n_{max}	Candidate motor maximum speed	3000	rpm	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	10 … 100000 rpm
SF	Safety margin on torque	1.25		assumption: Sizing convention.	1 … 3
$J_{ratio,max}$	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100

3. Assumptions

A1. Rigid transmission; the mechanism efficiency applies to the driving direction and is also used in the deceleration phase (conservative for peak, slightly pessimistic for RMS). (Servo-sizing practice.)
A2. The friction force is Coulomb friction on the moved mass; viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A3. The S-curve profile has sine-shaped acceleration over the same acceleration time, so its peak acceleration is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A4. mechanism (Mechanism) is taken as gearbox rotary. (Sizing convention.)
A5. J_{mech} (Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)) is taken

as 0 kg*m^2. (Sizing convention.)
A6. T_L (Constant load torque at the output (rotary)) is taken as 0 N*m. (Sizing convention.)
A7. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A8. SF (Safety margin on torque) is taken as 1.25. (Sizing convention.)
A9. $J_{ratio,max}$ (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

[20] k = 1 / i
= 1 / 10
= 0.1
Output radians per motor radian

[21] T_{load} = T_L / (i * η)
= 0 N·m / (10 × 0.85)
= 0 N·m
Load torque at the motor shaft

[22] J_{refl} = J_L / (i^2 * η)
= 0.5 kg·m² / (10^2 × 0.85)
= 0.0058824 kg·m²
Load inertia reflected to the motor shaft

[23] θ_m = s * 2 * pi * i
= 0.25 × 2 × π × 10
= 15.708 rad
Motor angle for the move

[24] ω_{pk} = θ_m / (t_{move} * (1 - f_{acc}))
= 15.708 rad / (0.3 s × (1 - 0.33333))
= 78.54 rad/s
Peak motor speed (trapezoid)

[25] α_{trap} = ω_{pk} / (f_{acc} * t_{move})
= 78.54 rad/s / (0.33333 × 0.3 s)
= 785.4 rad/s²
Trapezoidal angular acceleration at the motor

[27] α_{pk} = α_{trap}
= 785.4 rad/s²
= 785.4 rad/s²
Peak acceleration

[28] α_{pk} = α_{pk}
= 785.4 rad/s²
= 785.4 rad/s²
Peak motor angular acceleration

[29] J_{total} = J_{motor} + J_{mech} + J_{refl}
= 0.0005 kg·m² + 0 kg·m² + 0.0058824 kg·m²
= 0.0063824 kg·m²
Total inertia at the motor shaft

[30] T_{acc} = J_{total} * α_{pk}
= 0.0063824 kg·m² × 785.4 rad/s²
= 5.0127 N·m
Acceleration torque

[31] T_{pk} = T_{acc} + T_{load}
= 5.0127 N·m + 0 N·m
= 5.0127 N·m
Peak torque (accelerating against the load)

[32] T_{dec} = -T_{acc} + T_{load}
= −5.0127 N·m + 0 N·m
= -5.0127 N·m

Torque during deceleration

$$[33]\ n_{pk} = \omega_{pk}$$
$$= 78.54\ \text{rad/s}$$
$$= 750\ \text{rpm}$$

Peak motor speed

$$[34]\ P_{pk} = T_{pk} * \omega_{pk}$$
$$= 5.0127\ \text{N·m} \times 78.54\ \text{rad/s}$$
$$= 0.3937\ \text{kW}$$

Peak mechanical power

$$[35]\ t_{acc} = f_{acc} * t_{move}$$
$$= 0.33333 \times 0.3\ \text{s}$$
$$= 0.1\ \text{s}$$

Acceleration time

$$[36]\ t_{const} = t_{move} - 2 * t_{acc}$$
$$= 0.3\ \text{s} - 2 \times 0.1\ \text{s}$$
$$= 0.1\ \text{s}$$

Constant-velocity time

$$[37]\ t_{cycle} = t_{move} + t_{dwell}$$
$$= 0.3\ \text{s} + 0.7\ \text{s}$$
$$= 1\ \text{s}$$

Cycle time

$$[38]\ T_{rms} = \text{sqrt}((T_{pk}\text{^}2 * t_{acc} + T_{load}\text{^}2 * t_{const} + T_{dec}\text{^}2 * t_{acc}) / t_{cycle})$$
$$= \text{sqrt}(((5.0127\ \text{N·m})\text{^}2 \times 0.1\ \text{s} + (0\ \text{N·m})\text{^}2 \times 0.1\ \text{s} + (-5.0127\ \text{N·m})\text{^}2 \times 0.1\ \text{s}) / 1\ \text{s})$$
$$= 2.2417\ \text{N·m}$$

RMS torque over the cycle (thermal)

$$[39]\ J_{ratio} = (J_{refl} + J_{mech}) / J_{motor}$$
$$= (0.0058824\ \text{kg·m²} + 0\ \text{kg·m²}) / 0.0005\ \text{kg·m²}$$
$$= 11.765$$

Inertia ratio (load : motor)

$$[40]\ u_{peak} = SF * T_{pk} / T_{peak}$$
$$= 1.25 \times 5.0127\ \text{N·m} / 9\ \text{N·m}$$
$$= 0.69621$$

Peak torque utilisation with safety margin

$$[41]\ u_{rms} = SF * T_{rms} / T_{rated}$$
$$= 1.25 \times 2.2417\ \text{N·m} / 3\ \text{N·m}$$
$$= 0.93406$$

Continuous torque utilisation with safety margin

$$[42]\ u_{speed} = n_{pk} / n_{max}$$
$$= 750\ \text{rpm} / 3000\ \text{rpm}$$
$$= 0.25$$

Speed utilisation

$$[43]\ duty = T_{rms} / T_{rated}$$
$$= 2.2417\ \text{N·m} / 3\ \text{N·m}$$
$$= 0.74725$$

Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
k	0.1	Transmission constant
T_{load}	0 N·m	Load torque at motor
J_{refl}	0.0058824 kg·m²	Reflected load inertia
J_{total}	0.0063824 kg·m²	Total inertia

SYMBOL	VALUE	DESCRIPTION
θ_m	15.708 rad	Motor angle
a_{pk}	785.4 rad/s²	Peak acceleration
α_{pk}	785.4 rad/s²	Motor angular acceleration
T_{acc}	5.0127 N·m	Acceleration torque
T_{pk}	5.0127 N·m	Peak torque
T_{dec}	-5.0127 N·m	Deceleration torque
T_{rms}	2.2417 N·m	RMS torque
n_{pk}	750 rpm	Peak speed
P_{pk}	0.3937 kW	Peak power
J_{ratio}	11.765	Inertia ratio
duty	0.74725	Thermal duty
u_{peak}	0.69621	Peak utilisation
u_{rms}	0.93406	Continuous utilisation
u_{speed}	0.25	Speed utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque $SF \cdot T_{rms} \le T_{rated}$	u_{rms} = 0.93406	$u_{rms} \le 1$	0.934	
PASS	Peak torque $SF \cdot T_{pk} \le T_{peak}$	u_{peak} = 0.69621	$u_{peak} \le 1$	0.696	
PASS	Speed $n_{pk} \le n_{max}$	u_{speed} = 0.25	$u_{speed} \le 1$	0.25	

7. Applicability

WARNING — input.mu: μ (Friction coefficient (guide / bearing)) was not entered; the template default 0.1 is used.
WARNING — input.theta: θ (Incline from horizontal (+ moving uphill)) was not entered; the template default 0 deg is used.
WARNING — input.F_p: F_p (Process force opposing the move (linear)) was not entered; the template default 0 N is used.
WARNING — inertia-ratio: J_{motor}: the inertia ratio 11.8 exceeds 10.
WARNING — overhauling: T_{dec}: the motor brakes during deceleration (negative torque); the drive must absorb the regenerated energy.

8. Verification

```
Template elec.motor-sizing@1.0.1 Content hash 830a3b0397b596de Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Hughes A, Drury B. Electric Motors and Drives: Fundamentals, Types and Applications. 5th ed. Newnes; 2019.
2. Kollmorgen; Parker Hannifin. Servo motor sizing guides [manufacturer technical literature].